Local return moments and certified random-solving times for Rubik's Cube

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Abstract

Consider the random walk on the standard $3 \times 3 \times 3$ Rubik's Cube that independently chooses one of the eighteen face turns with equal probability. Let N denote the number of legal configurations and let T_e be the first hitting time of the solved configuration, including time zero. We give the certified bounds

$$1.0728316 < \frac{\mathbb{E}_\pi T_e}{N} < 1.0729459,$$

where π is uniform on the legal configurations. The same outward-rounded interval holds for $\mathbb{E}_x T_e/N$ for every configuration whose face-turn distance from solved is at least thirteen. The proof combines the established Green-function hitting-time identity with a degree-six polynomial resolvent certificate, exact return-word counts through length twelve, and a symmetry-averaged diameter estimate giving a spectral gap of at least $1/600$. The return counts were previously published in OEIS A061713 and are reproduced here by two independently implemented enumerations; all final inequalities use rational arithmetic. The certificate uses the classical matrix-moment framework to obtain an explicit finite-cube bound without computing the full spectrum or simulating rare solutions.

Keywords. Rubik's Cube; random walk; hitting time; Green's function; Kemeny's constant; spectral moments; polynomial certificate.

1 The problem and the result

A uniformly scrambled cube is solved with probability $1/N$ at every fixed time under a stationary random walk. This observation does not make its successive configurations independent, and it does not identify its expected first hitting time with N. A move can undo its predecessor, several moves on one face can cancel, and turns of opposite faces commute. Such relations create return correlations that remain relevant to the mean even though the state space is enormous.

We use a fixed reference orientation of the six centres. The letters U, D, R, L, F, B denote clockwise quarter turns of the up, down, right, left, front, and back faces, respectively, viewed from outside the face being turned. Let G be the group of legal cube configurations, let e denote its identity, and let

$$S = \{U, U^2, U^{-1}, D, D^2, D^{-1}, R, R^2, R^{-1}, L, L^2, L^{-1}, F, F^2, F^{-1}, B, B^2, B^{-1}\}.$$

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The inverse of a face turn rotates that face in the opposite direction, and its square is a half turn. We write $\operatorname{dist}(g, h)$ for the minimum number of moves from S needed to pass from g to h. The transition operator is

$$(Pf)(g) = \frac{1}{18} \sum_{s \in S} f(gs). \tag{1}$$

A half turn counts as one move, and immediate reversals are allowed. The order of the group is

$$N = 8!\, 3^7\, 12!\, 2^{10} = 43\,252\,003\,274\,489\,856\,000,$$

and its diameter in this metric is twenty [1, 2]. Define

$$T_e = \inf\{n \geq 0 : X_n = e\}. \tag{2}$$

All expected times below are measured in individual random face turns.

Theorem 1.1 (Certified cube bounds)**.** *For the walk* (1)*,*

$$1.0728316 < \frac{\mathbb{E}_\pi T_e}{N} < 1.0729459. \tag{3}$$

For every fixed $x \in G$ with $\operatorname{dist}(x, e) \geq 13$*,*

$$1.0728316 < \frac{\mathbb{E}_x T_e}{N} < 1.0729459. \tag{4}$$

Here π is uniform on all legal configurations, including e. The displayed decimal endpoints are rounded outwards from rational bounds.

In plain language, repeatedly choosing random face turns solves a Rubik's Cube after an average number of moves about seven point three percent greater than the number of its legal positions. This holds both when every starting position is equally likely and for every particular scramble that requires at least thirteen moves even with the best possible solution. Each random move is chosen independently, with all eighteen face turns equally likely, and a half turn counts as one move.

The first assertion places the correction to the state-count scale between 7.28316% and 7.29459%. In absolute units, the mean is approximately 4.6402×10^{19} to 4.6408×10^{19} random moves. The second assertion is a statement about each sufficiently distant starting configuration, not an assumption that a particular scramble is uniformly distributed.

The proof combines a finite list of closed-word counts with a bound on the nonconstant spectrum. The counts describe short returns to the starting configuration, while the spectral bound controls what remains beyond those local data. A polynomial brings the two inputs together within the established theory of Green functions, matrix moments, and inverse-form bounds [3, 4, 5, 6, 7, 8]. The constrained polynomial choice is the classical Christoffel-function minimization [9]. Related inverse-Laplacian bounds have been used to estimate pairwise commute times on graphs [10]. The return counts themselves were recorded by Healy [11]. Here these ingredients yield an explicit rational certificate for the cube and a bound applying separately to every sufficiently distant starting state.

Earlier work has considered random solving under other descriptions of the cube. Lee and Huang studied a corner-only model with a temperature-dependent transition rule [12]. Napier used sampled distance-layer transitions to approximate hitting times for combination puzzles, including the eighteen-move cube, while reporting failed consistency checks for the resulting $3 \times 3 \times 3$ estimate [13]. The present result bounds the specified walk on its full configuration group by exact arithmetic.

Research on cube scrambling concerns a different quantity: the time required for the distribution of configurations to approach uniformity [14]. Exact hitting-time formulae for other Cayley graph families are also an established line of work [15]. Our aim is narrower: an explicit, reproducible certificate for the mean accidental-solving time of the specified eighteen-move walk.

2 The Green quantity that must be bounded

The matrix P is symmetric, irreducible, and doubly stochastic. It is aperiodic because returns of both lengths two and three occur. Write $\Pi = \mathbf{1}\mathbf{1}^\top/N$ and define the centred Green matrix by

$$\mathcal{G} = (I - P + \Pi)^{-1} - \Pi. \tag{5}$$

Thus $\mathcal{G}\mathbf{1} = 0$ and $(I - P)\mathcal{G} = I - \Pi$. The matrix definition remains valid without aperiodicity; aperiodicity is needed for the ordinary convergent series

$$\mathcal{G} = \sum_{n\geq 0}(P^n - \Pi). \tag{6}$$

Proposition 2.1 (Established hitting-time identity)**.** *For the cube walk,*

$$\mathbb{E}_x T_e = N\big(\mathcal{G}(e,e) - \mathcal{G}(x,e)\big). \tag{7}$$

Consequently, with $p_n = P^n(e,e)$*,*

$$\kappa := \frac{\mathbb{E}_\pi T_e}{N} = \mathcal{G}(e,e) = \frac{1}{N}\operatorname{tr}\mathcal{G} = \sum_{n\geq 0}\left(p_n - \frac{1}{N}\right). \tag{8}$$

Proof. The right side of (7) vanishes at e. Away from e, applying $I - P$ gives one, so it solves the first-step equations for $\mathbb{E}_x T_e$. Their solution is unique on a finite irreducible chain. Averaging in x uses the zero column sums of $\mathcal{G}$. Left multiplication by a group element preserves P, and hence every matrix function of P has constant diagonal. This gives the trace equality. Equation (6) gives the final expression. These are standard Green-function identities; see [3, 4, 5]. □

If $1 = \lambda_1 > \lambda_2 \geq \cdots \geq \lambda_N$ are the eigenvalues of P, then

$$\kappa = \frac{1}{N}\sum_{j=2}^{N}\frac{1}{1-\lambda_j}. \tag{9}$$

Accordingly, $N\kappa$ is Kemeny's constant under the convention that a state's hitting time from itself is zero. It is not the mean positive return time. If $T_e^+ = \inf\{n \geq 1 : X_n = e\}$, the recurrence identity instead gives

$$\mathbb{E}_e T_e^+ = N. \tag{10}$$

The distinction is classical [3]. Conditioning a uniform initial state to be unsolved changes the mean in (8) to $N\kappa/(1 - 1/N)$, because the excluded initial state contributes zero.

3 A finite polynomial certificate

Let P more generally be a symmetric irreducible transition matrix on N vertices, invariant under a transitive group of vertex permutations. Its edges are the pairs with positive transition probability. For a polynomial b, set

$$\tau(b) = \frac{1}{N}\operatorname{tr} b(P).$$

Suppose the nonconstant eigenvalues are contained in $[\alpha, 1-\delta]$, where $\alpha < 1$ and $\delta > 0$. Put

$$c = \frac{1}{1-\alpha}, \qquad C = \frac{1}{\delta}. \tag{11}$$

Theorem 3.1 (Squared-polynomial resolvent certificate)**.** *Let q be a real polynomial of degree at most $m \geq 1$ with $q(1) = 1$. Define the polynomial*

$$a_q(z) = \frac{1 - q(z)^2}{1 - z}, \tag{12}$$

whose degree is at most $2m - 1$, and set

$$A_q = \tau(a_q) - \frac{a_q(1)}{N}, \qquad R_q = \tau(q^2) - \frac{1}{N}. \tag{13}$$

Then

$$A_q + cR_q \;\leq\; \frac{1}{N} \operatorname{tr} \mathcal{G} \;\leq\; A_q + CR_q. \tag{14}$$

In particular, the certificate uses only the moments $\tau(z^j)$ for $0 \leq j \leq 2m$ and the spectral enclosure.

Proof. For $z < 1$, the exact identity

$$\frac{1}{1 - z} = a_q(z) + \frac{q(z)^2}{1 - z} \tag{15}$$

implies

$$a_q(z) + cq(z)^2 \leq \frac{1}{1 - z} \leq a_q(z) + Cq(z)^2 \qquad (\alpha \leq z \leq 1 - \delta).$$

Sum over the nonconstant eigenvalues and divide by N. Since $q(1) = 1$, exclusion of the constant eigenvalue gives precisely the subtractions in (13). Equivalently, on $\mathbf{1}^\perp$ the scalar inequalities give the operator inequalities

$$a_q(P) + cq(P)^2 \;\preceq\; \mathcal{G} \;\preceq\; a_q(P) + Cq(P)^2. \tag{16}$$

□

The same certificate can be read as a residual bound for a positive definite inverse, as in classical matrix-moment methods [6, 7, 8]. On $\mathbf{1}^\perp$, put $B = I - P$, let $s(z) = (1 - q(z))/(1 - z)$, and set $y = s(P)u$ for $u \perp \mathbf{1}$. Its residual is $\rho = u - By = q(P)u$, and expansion gives

$$u^\top B^{-1} u = 2u^\top y - y^\top B y + \rho^\top B^{-1} \rho. \tag{17}$$

The first two terms equal $u^\top a_q(P) u$. Bounding the last term between $c \|\rho\|^2$ and $C \|\rho\|^2$ gives (16). The square therefore records the nonnegative residual energy, including when P has negative eigenvalues. Choosing $q(z) = z^m$ recovers the even-length partial Green sum and its spectral-gap tail bound. A polynomial adapted to the return moments can have a much smaller residual than that monomial.

Corollary 3.2 (Pointwise certificate beyond a finite ball)**.** *Under the hypotheses of Theorem 3.1, let e, x be vertices with $\operatorname{dist}(x, e) > 2m$. Then*

$$\tau(a_q) + c\tau(q^2) \;\leq\; \frac{\mathbb{E}_x T_e}{N} \;\leq\; \tau(a_q) + C\tau(q^2). \tag{18}$$

Proof. Let $v = \mathbf{e}_e - \mathbf{e}_x$ in the usual coordinate basis. Symmetry and the constant diagonal of $\mathcal{G}$ give

$$\frac{\mathbb{E}_x T_e}{N} = \frac{1}{2} v^\top \mathcal{G} v.$$

Since $v \perp \mathbf{1}$, apply (16). A polynomial of degree k in P has a zero (x, e) entry whenever $\operatorname{dist}(x, e) > k$. Both $a_q(P)$ and $q(P)^2$ therefore have zero (x, e) entries. Their diagonal entries equal their normalized traces, so the two quadratic-form bounds reduce to (18). □

The difference-vector inverse form also underlies quadrature bounds for graph commute times [10]. Here finite propagation removes the local off-diagonal terms whenever the starting state lies beyond the stated distance. This step uses no assumption about mixing within $2m$ steps. In the uniform certificate the constant eigenvalue is subtracted explicitly; in the pointwise proof it is annihilated by v.

4 Choosing the polynomial from exact moments

Let $A = 18P$ be the cube adjacency matrix and let w_j denote the number of length-j words in S equal to e. Then

$$p_j = \frac{w_j}{18^j}, \qquad \frac{1}{N} \operatorname{tr} A^j = w_j. \tag{19}$$

It is convenient to write $q(z) = r(18z)$, where

$$r(t) = \sum_{i=0}^{m} b_i t^i, \qquad r(18) = 1.$$

Define the integer Hankel matrix and the constraint vector by

$$H_m = (w_{i+j})_{i,j=0}^{m}, \qquad v_m = (1, 18, 18^2, \dots, 18^m)^\top. \tag{20}$$

Then $\tau(q^2) = b^\top H_m b$ and $v_m^\top b = 1$.

Proposition 4.1 (Minimum-width choice)**.** *If H_m is positive definite, the unique polynomial minimizing R_q is specified by*

$$b = \frac{H_m^{-1} v_m}{v_m^\top H_m^{-1} v_m}, \qquad \tau(q^2) = \frac{1}{v_m^\top H_m^{-1} v_m}. \tag{21}$$

When $C > c$, this is also the unique polynomial minimizing the width $(C - c)R_q$ of (14).

Proof. The subtraction $1/N$ in R_q is fixed under the constraint. Minimize $b^\top H_m b$ subject to $v_m^\top b = 1$, using a Lagrange multiplier or Cauchy–Schwarz in the H_m inner product. Strict positive definiteness gives uniqueness. □

For the normalized spectral measure of A, the minimum in (21) is the Christoffel function evaluated at 18, and the coefficient formula is its standard moment-matrix representation [9]. With $C > c$, it minimizes this certificate's width; it need not optimize each endpoint among all polynomial bounds. For the degrees used here, H_m is positive definite: it is the Gram matrix of $\mathbf{e}_e, A\mathbf{e}_e, \dots, A^m\mathbf{e}_e$. These vectors are linearly independent when there is a vertex at every distance through m, since a highest-degree nonzero term cannot cancel at a vertex of that distance. The cube has diameter twenty and is vertex-transitive, so this applies for $m \le 6$.

The construction in (21) can use rational linear algebra, but a final verifier need not invert a matrix. Given the rational coefficient vector, it suffices to check

$$v_m^\top b = 1, \qquad H_m b = (b^\top H_m b) v_m, \tag{22}$$

and evaluate the two endpoint expressions exactly. Positivity of the remainder follows from Theorem 3.1, not from a floating-point eigensolver.

5 A spectral enclosure from face subgroups and diameter

The moment construction leaves the spectral enclosure to be supplied. Diameter comparison bounds control the gap through paths joining pairs of states [3, Chapter 7, Section 7.1.9]. For the cube, we can also use the fact that the moves come from six four-element face subgroups. Averaging the paths over symmetries that permute these subgroups gives the following estimate.

Proposition 5.1 (Subgroup-symmetric diameter bound)**.** *Let a finite group be generated by subgroups $H_1, \dots, H_r$, each of order $h > 1$. Consider the walk that chooses i uniformly and then chooses uniformly from $H_i \setminus \{e\}$. Assume a group of automorphisms permutes the indexed subgroups transitively, and let D bound the word diameter for these moves. Then*

$$\lambda_{\min}(P) \geq -\frac{1}{h-1}, \qquad 1 - \lambda_2(P) \geq \frac{h}{2(h-1)D^2}. \tag{23}$$

Moves may be interpreted as a multiset if the subgroups overlap.

Proof. Work in the ℓ^2 space with uniform probability measure. Right translation R_s is unitary. The subgroup average

$$E_i = \frac{1}{h} \sum_{s \in H_i} R_s$$

is an orthogonal projection. Since

$$P = \frac{1}{r(h-1)} \sum_i (hE_i - I),$$

the lower spectral bound follows.

For a mean-zero f, put $D_s(f) = \|f - R_s f\|_2^2$. The projection identity gives

$$\sum_{s \in H_i \setminus \{e\}} D_s(f) = 2h \, \|(I - E_i)f\|_2^2 \,. \tag{24}$$

For every $s \in H_i$,

$$D_s(f) \leq 4 \, \|(I - E_i)f\|_2^2 = \frac{2}{h} \sum_{t \in H_i \setminus \{e\}} D_t(f). \tag{25}$$

The first inequality uses $R_s E_i = E_i$ and $\|I - R_s\| \leq 2$.

Choose, for each group element g, a word w_g of length $\ell_g \leq D$ representing it, with each letter assigned to its generating subgroup. Telescoping along the word and applying Cauchy–Schwarz yields

$$\|f - R_g f\|_2^2 \leq D \sum_{s \text{ in } w_g} D_s(f). \tag{26}$$

Average over the finite automorphism group in the hypothesis, applying an automorphism to every letter of every chosen word. For each fixed automorphism, its images of the endpoints still range uniformly over the group. By transitivity on the indexed subgroups, the expected number of letters assigned to each subgroup becomes $\mathbb{E}_g \ell_g / r \leq D/r$. Equations (25) and (26) give

$$2 \, \|f\|_2^2 = \mathbb{E}_g \, \|f - R_g f\|_2^2 \leq \frac{2D^2}{rh} \sum_{i=1}^{r} \sum_{s \in H_i \setminus \{e\}} D_s(f).$$

The Dirichlet form is

$$\langle f, (I - P)f \rangle = \frac{1}{2r(h-1)} \sum_i \sum_{s \in H_i \setminus \{e\}} D_s(f).$$

Combining the last two displays gives

$$\|f\|_2^2 \le \frac{2(h-1)D^2}{h}\langle f,(I-P)f\rangle.$$

The Rayleigh characterization proves the gap estimate. □

For the cube, take the six cyclic face subgroups, with $r = 6$ and $h = 4$. A proper spatial rotation acts by conjugation on the facelet permutations, normalizes the fixed-centre cube group, and permutes the face subgroups. These conjugations act transitively on the six subgroups, so they provide the automorphisms required in the proof. The reference orientation and move set of the walk remain fixed. The established diameter bound $D = 20$ [1] gives

$$\operatorname{spec}(P)\setminus\{1\} \subseteq \left[-\frac{1}{3},\, 1-\frac{1}{600}\right]. \tag{27}$$

We may therefore take

$$c = \frac{3}{4}, \qquad C = 600. \tag{28}$$

This enclosure supplies the global control required by the moment certificate; its gap bound need not be sharp.

6 Exact return words and computational verification

The necessary local data are shown in Table 1. All listed values agree with OEIS A061713, recorded by Alexander D. Healy in 2001 [11]. They are not new sequence entries.

Table 1: Closed words for the eighteen-move walk. Every entry was reproduced by both the cubie and sticker enumerations.

n	w_n	n	w_n
0	1	7	312 480
1	0	8	3 148 032
2	18	9	27 073 152
3	36	10	261 446 688
4	720	11	2 407 791 936
5	3 600	12	23 168 736 768
6	42 624		

Let $c_k(g)$ count the length-k words with value g. Starting with $c_0(e) = 1$, the recurrence is

$$c_{k+1}(g) = \sum_{s\in S} c_k(gs^{-1}). \tag{29}$$

Because $S = S^{-1}$, reversing and inverting a word shows that $c_k(g) = c_k(g^{-1})$. Hence

$$w_{2k} = \sum_g c_k(g)^2, \qquad w_{2k-1} = \sum_g c_k(g)c_{k-1}(g). \tag{30}$$

Thus moments through order twelve require only distributions through six moves, not an enumeration of every length-twelve word.

The largest stored distribution has 8 240 087 configurations and total weight $18^6 = 34\,012\,224$. Its support size agrees with the sum of the known distance counts through six [2]. The C++ implementation represents corner and edge identities and orientations by

two integer words. A hash table indexes these states, but equality compares the full keys: hash collisions are not discarded or treated probabilistically. Word counts and their inner products use integer arithmetic. The crude bound $w_{12} \le 18^{12} < 2^{64}$ also bounds every nonnegative partial sum used here.

Move tables are generated from exact rotations of integer coordinate triples. Tests check face-turn orders, inverses, powers, and commuting opposite-face turns. A second representation tracks all forty-eight moving stickers and agrees with the cubie representation on 1,000 deterministic test-word steps. In addition, an independently implemented enumeration constructs its own sticker permutations from integer rotation matrices and uses sorting of complete sticker states. It reproduces every return count through order twelve without using the cubie move tables. Further checks verify the conversion between both representations and the corner and edge orientation constraints.

The final certificate has a separate verifier based only on Python's standard-library rational arithmetic. It checks the polynomial identity, the moment normal equations, agreement with the stored return counts, the outward rounding of the uniform and pointwise theorem bounds, and the degree-three through degree-six intervals in Table 3. The supplied source separates generation, enumeration, polynomial construction, and verification.

7 The explicit certificate and proof of the cube theorem

Take $m = 6$. The polynomial from (21) is $q(z) = r(18z)$, where

$$r(t) = \frac{1}{B}\sum_{j=0}^{6} \beta_j t^j, \qquad B = 147677640811420994309176, \tag{31}$$

and the integer numerators are given in Table 2.

Table 2: Integer numerators of the rational certificate (31).

j	β_j
0	−17977105357205381336
1	−49385224724300216876
2	6297242415075501624
3	5663254501173955257
4	−487257039354164436
5	−125614143990543078
6	11819953525051755

Define three rational constants

$$C_0 = \frac{731157667743011107668024391505232221547122643}{681521424863346113452159814217869705546181218}, \tag{32}$$

$$R_0 = \frac{7033126817141887}{36919410202855248577294}, \tag{33}$$

$$F_1 = \frac{280757476890446720214528}{18459705101427624288647}. \tag{34}$$

Direct rational evaluation using Table 1 gives

$$r(18) = 1, \qquad \tau(a_q) = C_0, \qquad \tau(q^2) = R_0, \qquad a_q(1) = F_1. \tag{35}$$

For orientation, their approximate sizes are

$$C_0 \approx 1.07283152233932734368, \qquad R_0 \approx 1.9049943589288335 \times 10^{-7}.$$

Proof of Theorem 1.1. Apply Theorem 3.1 with (28) and (35). The exact uniform-initial bounds are

$$L = C_0 + \frac{3}{4}R_0 - \frac{F_1 + 3/4}{N}, \tag{36}$$

$$U = C_0 + 600R_0 - \frac{F_1 + 600}{N}. \tag{37}$$

Rational comparison gives

$$L > 1.0728316, \qquad U < 1.0729459.$$

Since $2m = 12$, Corollary 3.2 gives, whenever $\operatorname{dist}(x, e) \geq 13$,

$$C_0 + \frac{3}{4}R_0 \leq \frac{\mathbb{E}_x T_e}{N} \leq C_0 + 600R_0. \tag{38}$$

Rational comparison places these two endpoints strictly inside the same outward-rounded interval. All the required data and the coefficient polynomial appear above, so these comparisons are finite arithmetic checks. □

More decimal places of the uniform endpoints are

$$L = 1.072831665213904262974\ldots,$$
$$U = 1.072945822000863059468\ldots.$$

Their difference is approximately $1.1415678696 \times 10^{-4}$, less than 0.0107% of the enclosed normalized mean. The pointwise endpoints differ from the corresponding uniform endpoints only by the explicitly retained terms in (36)–(37); neither calculation replaces N by infinity.

Table 3 shows the effect of increasing the polynomial degree while retaining the same global spectral enclosure. Its endpoints are rounded outwards. Already-known local moments can therefore strengthen the certificate without any new estimate of the cube's entire spectrum.

Table 3: Certified uniform-initial intervals from successively more return moments.

Degree m	Highest moment	Lower endpoint	Upper endpoint
3	6	1.0725423	1.2788134
4	8	1.0728083	1.0892097
5	10	1.0728298	1.0742176
6	12	1.0728316	1.0729459

The pointwise assertion covers all but a small ball. Summing the published exact distance counts through twelve [2] gives

$$|B_{12}(e)| = 43\,689\,000\,394\,782, \qquad \frac{|B_{12}(e)|}{N} \approx 1.010104 \times 10^{-6}.$$

Thus the pointwise interval applies to more than 99.99989% of legal configurations. This count describes which starting vertices satisfy the theorem; it is not an estimate of a hitting probability or a waiting-time quantile.

8 Interpretation, limitations, and scope of the contribution

The correction factor has a local origin but requires global control. Local return words determine a finite list of moments, and those moments produce a polynomial that is small in mean square over the cube spectrum while remaining one at the stationary eigenvalue. The diameter estimate prevents the remaining nonconstant spectrum from approaching that eigenvalue arbitrarily closely. Together these facts bound the inverse Laplacian without computing its full matrix.

The graph's short optimal solution paths and its enormous random hitting times are not in conflict. Diameter enters this proof only as a coarse control on slow spectral modes. It supplies no preferred direction to a random move. Nor does the result contradict (10): the positive return time from solved and the first hitting time from a uniform initial state are different random variables.

The narrow interval is not asserted for every nearby configuration. At e, the hitting time is zero by definition. Other configurations within twelve moves can have nonzero off-diagonal entries in the two matrix polynomials used in Corollary 3.2. Their certificate would require local transition moments to the target, not only return moments. The pointwise proof identifies exactly where that additional information enters.

The present theorem concerns the specified independent eighteen-move walk. Quarter-turn-only walks, rules that forbid an immediate reversal, and random choices of longer algorithms define different processes. Their local moments and spectral enclosures must be reconsidered. In particular, a rule depending on the previous move is not a Markov chain on cube configurations alone without an appropriate state augmentation.

A mean also does not determine the distribution of T_e. Exponential approximations of stationary hitting times have a substantial theory, including recent refinements for vertex-transitive graphs [16]. We do not claim a new distributional theorem here. The result supplies an explicit scale that such a theorem could use after its hypotheses and discrete-time conventions are checked.

The contribution is the explicit rational cube certificate, together with the subgroup-symmetric gap estimate used to close it and the finite-propagation application to every state outside the radius-twelve ball. The Green identities, residual and moment methods, Christoffel minimization, and integer return data belong to the prior work cited above.

Reproducibility

The accompanying source package contains this article's LaTeX source, exact count and certificate files, the move generator, both C++ enumeration programs, a rational polynomial constructor, and a separate standard-library verifier. Its README gives the commands and dependencies needed to reproduce the calculation. The full cube diameter computation is an external established input [1]. The local enumerations and arithmetic checks described in Section 6 were executed during preparation; the diameter computation was not rerun.

AI assistance

AI tools assisted the formulation, literature comparison, proof development, and preparation of this manuscript within the Mathematics Knowledge and Connections Program.

References

[1] T. Rokicki, H. Kociemba, M. Davidson, and J. Dethridge. The diameter of the Rubik's Cube group is twenty. *SIAM Journal on Discrete Mathematics* **27**(2) (2013), 1082–1105. doi:10.1137/120867366.

[2] T. Rokicki, H. Kociemba, M. Davidson, and J. Dethridge. *God's Number is 20*. Project website and distance-count table. `https://www.cube20.org/`. Accessed 3 October 2026.

[3] D. Aldous and J. A. Fill. *Reversible Markov Chains and Random Walks on Graphs*. Unfinished monograph, 2002; recompiled 2014. See Chapter 2, Section 2.2. `https://www.stat.berkeley.edu/~aldous/RWG/book.html`.

[4] F. Chung and S.-T. Yau. Discrete Green's functions. *Journal of Combinatorial Theory, Series A* **91**(1–2) (2000), 191–214. doi:10.1006/jcta.2000.3094.

[5] A. Beveridge. A hitting time formula for the discrete Green's function. *Combinatorics, Probability and Computing* **25**(3) (2016), 362–379. doi:10.1017/S0963548315000152.

[6] G. H. Golub and G. Meurant. Matrices, moments and quadrature II; how to compute the norm of the error in iterative methods. *BIT Numerical Mathematics* **37**(3) (1997), 687–705. doi:10.1007/BF02510247.

[7] G. H. Golub and G. Meurant. *Matrices, Moments and Quadrature with Applications*. Princeton University Press, 2010. doi:10.1515/9781400833887.

[8] C. Li, S. Sra, and S. Jegelka. Gaussian quadrature for matrix inverse forms with applications. *Proceedings of Machine Learning Research* **48** (2016), 1766–1775. `https://proceedings.mlr.press/v48/lig16.html`.

[9] J. B. Lasserre and E. Pauwels. The empirical Christoffel function with applications in data analysis. *Advances in Computational Mathematics* **45**(3) (2019), 1439–1468. doi:10.1007/s10444-019-09673-1.

[10] F. Bonchi, P. Esfandiar, D. F. Gleich, C. Greif, and L. V. S. Lakshmanan. Fast matrix computations for pairwise and columnwise commute times and Katz scores. *Internet Mathematics* **8**(1–2) (2012), 73–112. doi:10.1080/15427951.2012.625256.

[11] A. D. Healy. Number of closed walks of length n on a $3 \times 3 \times 3$ Rubik's Cube. *The On-Line Encyclopedia of Integer Sequences*, A061713, 21 June 2001. `https://oeis.org/A061713`. Accessed 3 October 2026.

[12] C.-L. Lee and M.-C. Huang. The Rubik's cube problem revisited: a statistical thermodynamic approach. *The European Physical Journal B* **64**(2) (2008), 257–261. doi:10.1140/epjb/e2008-00301-0.

[13] J. O. H. Napier. *Connectivity of Combination Puzzles: Using Machine Learning on Groups*. M.S. thesis, Eastern Michigan University, 2021. Author manuscript.

[14] Y. Qu, T. Rokicki, and H. Yang. Rubik's Cube scrambling requires at least 26 random moves. Preprint, arXiv:2410.20630v1 (2024). `https://arxiv.org/abs/2410.20630`.

[15] X.-M. Zhu, X.-D. Jiang, and X. Yang. Exact hitting time formulas for Cayley graphs. *Linear Algebra and its Applications* **743** (2026), 56–75. doi:10.1016/j.laa.2026.04.018.

[16] N. Berestycki, J. Hermon, and L. Teyssier. Stationary hitting times on vertex-transitive graphs. Preprint, arXiv:2601.03864v1 (2026). `https://arxiv.org/abs/2601.03864`.